



Julie Kallini

Ph.D. Student | NLP | Computational Linguistics

-  juliekallini.com
-  kallini@stanford.edu
-  [juliekallini](#)
-  [jkallini](#)

About Me

I am a second-year Computer Science Ph.D. student at Stanford University, where I am a member of the Natural Language Processing Group. Previously, I worked for nearly two years as a software engineer at Meta, where I used machine learning for privacy problems in advertisements. Before joining Meta, I graduated from Princeton University, where I was fortunate to conduct computational linguistics research. I am originally from Los Angeles, California.

Research Interests

My work lies at the intersection of NLP, machine learning, and linguistics. I am particularly interested in tokenization and vocabulary, with the goal of improving language models' efficiency and fairness across languages. I am also interested in understanding what deep learning can reveal about linguistics and language acquisition.

Education

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| 2023—Pres | Ph.D. in Computer Science
• GPA: 4.21/4.00
• Advisors: Christopher Potts and Dan Jurafsky | Stanford University |
| 2017—2021 | B.S.E. in Computer Science, Minor in Linguistics
• GPA: 3.94/4.00 (<i>summa cum laude</i>)
• Advisor: Christiane Fellbaum | Princeton University |

Papers

- [1] **Julie Kallini**, Shikhar Murty, Christopher D. Manning, Christopher Potts, and Róbert Csordás. **MrT5: Dynamic Token Merging for Efficient Byte-level Language Models**. October 2024.
- [2] **Julie Kallini**, Isabel Papadimitriou, Richard Futrell, Kyle Mahowald, and Christopher Potts. **Mission: Impossible Language Models**. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, Bangkok, Thailand, August 2024. **Best Paper Award**.
- [3] **Julie Kallini** and Christiane Fellbaum. **What to Make of make? Sense Distinctions for Light Verbs**. In *Global WordNet Conference 2023*, Donostia-San Sebastian, Spain, January 2023.
- [4] **Julie Kallini** and Christiane Fellbaum. **Computational Approaches for Understanding Semantic Constraints on Two-termed Coordination Structures**. In *Text, Speech, and Dialogue*, Brno, Czech Republic, 2022.
- [5] **Julie Kallini** and Christiane Fellbaum. **A Corpus-based Syntactic Analysis of Two-termed Unlike Coordination**. In *Findings of the Association for Computational Linguistics: EMNLP 2021*, Punta Cana, Dominican Republic, November 2021.

Teaching Experience

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| 2022—Pres | Instructor, Applied Machine Learning
Built scikit-learn projects and designed course materials for introductory ML course for working professionals; taught at weekly lectures and office hours. | Uplimit (formerly CoRise) |
| 2021 | Head TA, Machine Translation
Designed coding exercises in Python and led weekly classes on NLP topics. | Princeton University |
| 2020 | TA, Computer Networks
Taught course material and ran weekly in-class discussion sessions. | Princeton University |
| 2018—2019 | Lead Grader/CA, Algorithms and Data Structures
Designed Java assignment rubrics; trained and supervised other graders. | Princeton University |

Industry Experience

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| 2021—2023 | Software Engineer, Machine Learning for Ads Privacy
Built classifiers to detect sensitive topics in media on the Meta family of apps. | Meta |
| 2020 | Software Engineer Intern, Community Integrity
Designed data analysis and monitoring tools for content moderation. | Meta |
| 2019 | Software Engineer Intern, Android Development
Created an Android app that organizes and streamlines office hours queues. | Meta |

Fellowships, Awards, and Honors

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| 2024 | Best Paper Award (for Mission: Impossible Language Models) | ACL 2024 |
| 2023 | NSF Graduate Research Fellowship | Stanford University |
| 2023 | School of Engineering Graduate Fellowship | Stanford University |
| 2023 | EDGE Fellowship | Stanford University |
| 2021 | Phi Beta Kappa | Princeton University |
| 2021 | Phillip Goldman '86 Senior Prize (Top CS Academic Prize) | Princeton University |
| 2021 | Outstanding CS Senior Thesis Prize | Princeton University |
| 2021 | Outstanding Student Teaching Award | Princeton University |
| 2019 | Tau Beta Pi (Engineering Honor Society) | Princeton University |